

## IS 470/570 – Business Intelligence - Assignment 2

**Assignment Name:** Voting Behavior

---

**Purpose:** This assignment will help you understand how machine learning (ML) models such as Decision Tree (DT), Naïve Bayes (NB), and K-Nearest Neighbor (KNN), are used to solve real-world business problems such as predicting voting behavior. You'll explore a dataset, build classification models, and evaluate their performance. This foundational project builds your skills for future assignments and gives you hands-on experience relevant to careers in data science, analytics, and business intelligence.

---

**Skills:** By completing this assignment, you will practice and strengthen the following skills:

- Using Google Colab for data analysis and model building
  - Applying the CRISP-DM process to perform a data mining task
  - Visualizing and interpreting data
  - Building and evaluating classification models
  - Communicating findings effectively in a structured report
- 

**Knowledge:** You'll become familiar with:

- Basic principles of voting behavior
  - Key performance metrics in classification tasks (accuracy, precision, recall, f1-score)
  - Overfitting and generalization
  - How to use ML models for classification problems
  - Common challenges in data preprocessing (e.g., missing values, outliers, unbalanced data)
- 

**Task:**

1. **Setup:**
  - Open Google Colab and prepare your environment.
  - Download the dataset `Gaming Ballot Dataset.xls` from Canvas and upload it into your drive and notebook.
  - Use code cells for code only and add your explanations in text cells.
  - This should look like a project report, so include sections and subsections as needed in an organized way.

- A Gaming Ballot Data Description.pdf file is provided with a short definition of variables.
- To explore more, search online to understand the meaning of each variable/feature (columns) in the dataset, if needed.

## 2. CRISP-DM Process (the sections in your report):

- **Business Understanding:**
  - Briefly describe the context of voting behavior in general and how the results can impact related industries; define the voting behavior problem based on your understanding.
  - Explain which industries can benefit from having a predictive model and how?.
- **Data Understanding:**
  - What is each variable's definition?
  - How do these variables relate to the context of the problem in hand?
  - Explore the dataset using summary statistics and charts (include at least 10 charts that are interesting to you). Consider variables separately and in relation to each other. After each chart/statistics explain your findings or insights about the variable(s).
  - What is your target variable? What is its type?
  - Do you think other existing variables can help build a meaningful and predictive classification model for this target variable? Can we use all of them as predictors?

Note: other than highly correlated variables and IDs, be mindful of cheating variables and **data leakage**. Data leakage happens when information that would not be available at prediction time “leaks” into the model.

- **Data Preparation:**
  - Fix the variables' types (if needed).
  - Explain and handle missing values, outliers, highly correlated variables, and cheating variables.
  - Comment on whether the data is balanced. If not, explain what techniques could be used to fix it (you don't have to fix it in this assignment).
  - Perform dummy coding for the categorical variables.
- **Modeling:**
  - Build a decision tree classifier.
    1. Identify and explain key variables influencing the prediction.
    2. Try to enhance the model's predictive performance by changing the depth.
    3. Examine the models' performance on the **training** dataset. Identify which max\_depth leads to underfitting, which leads to overfitting, and which provides the best generalization to the test data. (max\_depth is a hyperparameter that can be tuned to improve the model's performance).

- Try KNN.
  1. Try to enhance the model's predictive performance by changing K.
  2. Examine the models' performance on the **training** dataset. Identify which K leads to underfitting, which leads to overfitting, and which provides the best generalization to the test data. (K, the number of neighbors) is a hyperparameter that can be tuned to improve the model's performance).
- Build an NB model. Can we improve this model's performance? Does it have any hyperparameters?
- In one table, compare the best performing DT, KNN, and NB with respect to model's performance metrics: accuracy, precision, recall, F1-score, confusion matrix. How good these models are? Choose your overall best performing model.
- **Evaluation:**
  - Explain how your top model can be used to address the problem we started from: voting behavior?
  - Discuss whether your model is practical for businesses.

### Criteria for Success:

| Category                                | Points | What Success Looks Like                                                                                                                                                                            |
|-----------------------------------------|--------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Discussions/<br>Explanations            | 10     | Thoughtful reflection on the problem, data (variables), findings from data, model limitations, and how the results could be applied in real life.                                                  |
| Data Understanding<br>and Preprocessing | 10     | Understanding the variables in the dataset using their definitions, statistics, chart, Proper handling of missing values, outliers, and any anomalies.                                             |
| Visualization                           | 10     | 10 <b>insightful</b> charts that explore variable distributions and relationships. Well-labeled and explained.                                                                                     |
| Results & Model<br>Evaluation           | 10     | Models built correctly. Training performance and generalization is discussed. Evaluation metrics reported and explained clearly. Includes discussion of variable importance and model improvement. |
| Presentation &<br>Organization          | 10     | Notebook is clear, well-organized, with labeled sections and explanations. All questions are answered. Code is readable, and output is clean.                                                      |

- ✓ **Submission Format:** HTML file
- ✓ **File Naming:** LastnameFirstname.html
- ✓ **Total Points Available:** 50

Good Luck!